

ATE Social Survey

Technical Guide

Community attitude, knowledge, and sentiment analysis towards elephants

Generated September 23, 2026

Workflow id: **social_survey**

1. Overview

The **social_survey** workflow processes Amboseli Trust for Elephants (ATE) community questionnaire responses recorded as **atequestionnaire_rep** events in EarthRanger. It cleans and relabels the survey answers, produces demographic, attitude, and knowledge charts, computes a per-respondent elephant sentiment score, compares sentiment across demographic groups, maps respondents over Amboseli context layers, and assembles everything into an interactive dashboard and a Word report.

This guide is written directly from **spec.yaml**. The workflow is compiled with **wt-compiler** (see **pixi.toml**) into the **ecoscope-workflows-social-survey-workflow/** package.

Output summary

Output type	Count	Description
Dashboard	1	53 widgets: 50 charts and 3 maps (gather_dashboard)
Pie charts	11	Demographics, threats, feelings about wildlife, elephant encounters
Bar charts (binned)	9	Age, household size, livestock counts, land owned and farmed
Bar charts (Yes/No)	10	Behaviour and mitigation questions
Bar charts (True/False)	8	Elephant knowledge questions
Likert chart	1	9 agree/disagree attitude statements
Boxplots	5	Sentiment by gender, age group, education, tribe, marital status
Tukey HSD plots	4	Pairwise sentiment comparisons by gender, marital status, age group, education
Scatter plots (OLS)	2	Sentiment against household size and age
Maps	3	Survey locations, gender, attitude category
Data files	3	survey-events.gpkg, demographic-data.parquet, elephant-sentiment-table.gpkg
Word report	1	social_survey_report.docx

2. Dependencies & Configuration

2.1 User-configurable form

Only three sections appear in the run form; everything else is fixed in **spec.yaml** via **partial** arguments.

Section	Task	Fields
Workflow Details	set_workflow_details	Workflow Name (required), Workflow Description
Time Range	set_time_range	Since, Until (required), Timezone, Time Format (advanced; default %d %b %Y %H:%M:%S)
Connect to EarthRanger	set_er_connection	Data Source

2.2 Groupers

set_groupers is fixed to **groupers: []**. All survey records are processed as a single dataset, with no per-group split. The empty grouper list is passed through to the dashboard only.

2.3 Base maps

Two tile layers are fixed via **set_base_maps_pydeck**:

Layer	URL (abbreviated)	Opacity
ArcGIS World Hillshade	server.arcgisonline.com/.../Elevation/World_Hillshade/MapServer/tile/{z}/{y}/{x}	1.0
ArcGIS World Street Map	server.arcgisonline.com/.../World_Street_Map/MapServer/tile/{z}/{y}/{x}	0.15

Both layers use max_zoom: 20.

2.4 Files downloaded at runtime

Four files are downloaded from Dropbox with **ecoscope_workflows_ext_ste.tasks.io.fetch_and_persist_file** (overwrite_existing: false, retries: 3, unzip: false) into **ECOSCOPE_WORKFLOWS_RESULTS**:

Task id	File	Purpose
ambo_ranch_layers	amboseli_ranch_conservancies_layer.s.gpkg	Land-use zones, coloured by the land_use column
ambo_bound_fence	amboseli_group_ranch_boundaries_x_electric_fence.gpkg	Filtered into two layers: land_use == 'Ranch boundaries' and land_use == 'Electric Fence'
ambo_conservancies	amboseli_group_ranch_boundaries.gpkg	Loaded and reprojected. Not currently drawn on any map
download_word_template	ate_survey_template_280857_update.d.docx	docxtpl/Jinja Word template for the report

3. Survey Data Pipeline

3.1 Event retrieval

Parameter	Value
Task	get_events
Event type	atequestionnaire_rep
Columns retained	id, time, event_type, event_category, reported_by, serial_number, geometry, created_at, event_details
raise_on_empty	true (the run fails if no events are found)
include_details	true
include_display_values	true
force_point_geometry	true
include_null_geometry	false
include_updates include_related_events	/ false / false

3.2 Flattening and renaming

Step (task id)	Task	What it does
extract_event_date	extract_column_as_type	Adds a date column from time
process_survey_details	process_events_details (ext-custom)	Replaces event_details choice values with their EarthRanger display titles (map_to_titles: true, ordered: true)
normalize_survey_event s	normalize_json_column	Flattens event_details into event_details__* columns (skip_if_not_exists, sort_columns)
drop_survey_prefix	drop_column_prefix (ext-custom)	Strips the event_details__ prefix (duplicate_strategy: keep_original)
map_survey_columns	map_columns	Drops event_type and event_type_display; renames about 80 bilingual question headings to English only (e.g. <i>What is your Age? (Kaja larin liata?)</i> → <i>What is your Age</i>). raise_if_not_found: false

Note: Because **raise_if_not_found** is false, a question whose heading changes in EarthRanger won't fail the run. Its column simply keeps the original bilingual heading, and any downstream chart that expects the English name is skipped.

3.3 Record exclusion

Rows are filtered with **ecoscope_workflows_ext_ste.tasks.filter.filter_rows** (reset_index: true), in this order:

Task id	Column	Rule	Keeps
exclude_invalid_serial	serial_number	ne 46283	Everything except the known invalid entry
exclude_invalid_age	What is your Age	gt 18	Respondents older than 18
exclude_invalid_household	How many People are there in your Household	gt 1	Households of 2 or more people

Note: These filters run before missing values are filled, so rows with a blank age or household size are also dropped.

3.4 Missing values and type conversion

Task id	Task	Columns	Behaviour
fill_missing_columns	fill_missing_values (ext-mnc)	11 numeric columns: livestock counts (cattle, dogs, donkeys, sheep/goats, other animals), household size, land owned and farmed, age, participant_num_cows, participant_num_shoats	fill 0
convert_cols_numeric	convert_column_values_to_numeric	Same 11	to numeric
fill_string_columns	fill_missing_values (ext-mnc)	About 75 text/choice columns	fill 'No Response'
convert_num_int	convert_columns_to_int (ext-mnc)	Same 11 numeric	errors: coerce, fill_value: 0

3.5 Answer relabelling

A chain of **replace_column_values** tasks (inplace: true, errors: ignore) strips the Maa translation from each answer and merges variants:

Task id	Columns	Mapping (raw → label)
replace_yes_no	11 Yes/No questions	Yes (ee)→Yes, No (aa)→No, I Don't Know (Mayolo)→I Don't Know; No Response and Not Applicable unchanged
replace_true_false	7 knowledge statements	True, False, I Don't Know, No Response (unchanged)
replace_agree_disagree	11 attitude statements	Strongly Agree (tipat ake)→Strongly Agree, Agree (tipat)→Agree, Neutral (tipat nemetipat)→Neutral, Disagree (meetipat)→Disagree, Strongly Disagree (meetipat pi)→Strongly Disagree, I Don't Know (mayolo)→I Don't Know
replace_gender_participant	Gender of Participant	Unknown→No Response
replace_ele_area	Having Elephants in this Area is	Very Good / Good / Neutral / Bad / Very Bad / I Don't Know / Prefer not to Answer
replace_how_often	How often seen elephants in the last year	Every Day / Week / Month / Few Months / Year, Never, I Don't Know

Task id	Columns	Mapping (raw → label)
replace_overall_wildlife	Overall feelings about wildlife	Strongly Like / Like / Neutral / Dislike / Strongly Dislike, I Don't Know, Prefer not to Answer
replace_age_groups	What is your Age Group	Senior Elder, Junior Elder, Elder → Elder; Moran, Kidemi Mamas unchanged
replace_highest_level	Highest level of completed education	None→No Response; Primary, Secondary, University, Post Grad unchanged
replace_marital_status	What is your marital status	About 90 free-text spellings (English and Maa: iama, iamishe, eamishe, itu ...) → Married, Single, Divorced, or No Response. Widowed → Single; Separated → Married

The cleaned table is saved as **survey-events.gpkg** (task events_gpkg).

4. Pie Charts

Task: `ecoscope_workflows_ext_wwf_virunga.tasks.plot._plot.draw_pie_chart`, one per question, with `value_column = label_column = the question`. `textinfo: percent+label+value`, `font_size 10`, legend shown. Each is saved with `persist_text`.

Widget	Question	Output file (.html / .png)
0	What is your Age Group	what_is_your_age_group_pie_chart
1	What is your Arrangement for the Land that You are Farming On	what_is_the_land_arrangement_pie_chart
2	What is your Highest Level of Completed Education	highest_level_of_education_pie_chart
3	What is your Tribe	what_is_your_tribe_pie_chart
4	Greatest Threat to Successful Crop Production	greatest_threat_to_crop_production_pie_chart
5	Greatest Threat to your Livestock	greatest_threat_to_livestock_pie_chart
6	What is your marital status	what_is_your_marital_status_pie_chart
7	Overall feelings about wildlife	overall_feeling_about_wildlife_pie_chart
8	How often seen elephants in the last year	how_often_have_you_seen_elephants_in_the_last_year_pie_chart
9	How do you feel when an elephant is speared or poisoned in retaliation	how_do_you_feel_witnessing_elephant_harmed_pie_chart
10	Gender of Participant	gender_of_participant_pie_chart

5. Binning & Bar Charts

5.1 Numeric binning

Nine chained **apply_classification** tasks add a bin column for each numeric answer using **natural_breaks**, **k = 5**. **order_categorical_by_number** then orders every bin column numerically so the bars sort correctly.

Input column	Bin column	Labels
What is your Age	age_bins	ranges, 0 dp
How many Cattle does Your Household Own	cattle_bins	ranges, 0 dp
How many Dogs does Your Household Own	dog_bins	fixed: 0, 0 - 1, 1 - 2, 2 - 4, 4 - 8
How many Donkey does Your Household Own	donkey_bins	fixed: 0, 0 - 3, 3 - 7, 7 - 25, 25 - 56
How many Other Animals does Your Household Own	other_animals_bins	ranges, 0 dp
How many People are there in your Household	other_people_bins	ranges, 0 dp
How many Sheep/Goats does Your Household Own	sheep_goats_bins	ranges, 0 dp
How much Land do you Grow Crops on each Season (in Acres)	land_grown_crops_bins	ranges, 0 dp
How much Land do you Own (in Acres)	land_owned_bins	ranges, 0 dp

Note: The dog and donkey labels are hard-coded from the current data distribution. For those two charts, the '0' bin is filtered out (`filter_df ne '0'`) and the column is converted back to string so the unused category disappears.

5.2 Bar chart settings

Task: **ecoscope_workflows_ext_wwf_virunga.tasks.plot.draw_bar_chart**. Every bar chart counts respondents (column id, `agg_func`: count) per category, in grouped mode, with marker colour #6495ed, labels shown, background #f5f5f5, and no legend.

5.3 Bar chart inventory

Widget	Source	Category	Output file
11	binned	age_bins	age_bins_bar_chart
12	binned	cattle_bins	cattle_bins_bar_chart
13	binned	dog_bins (non-zero)	dog_bins_bar_chart
14	binned	donkey_bins (non-zero)	donkey_bins_bar_chart
15	binned	other_animals_bins	other_animal_bins_bar_chart
16	binned	other_people_bins	other_people_bins_bar_chart

Widget	Source	Category	Output file
17	binned	sheep_goats_bins	sheep_goats_bins_bar_chart
18	binned	land_grown_crops_bins	land_grown_crops_bins_bar_chart
19	binned	land_owned_bins	land_owned_bins_bar_chart
20	Yes/No	Benefit from the KCCDT Big Life Electric Fence	do_you_benefit_from_electric_fence_bar_chart
21	Yes/No	Choose routes/schedules/water based on elephant presence	activities_altered_elephant_presence_bar_chart
22	Yes/No	Protect where you get water from elephants	protect_water_from_elephants_bar_chart
23	Yes/No	See more elephants now compared to the past	more_eles_past_bar_chart
24	Yes/No	Measures to prevent crop damage	family_protection_eles_bar_chart
25	Yes/No	Measures to prevent injuring livestock	family_protection_eles_livestock_bar_chart
26	Yes/No	Signs of illness in wild animals	illness_wild_bar_chart
27	Yes/No	Involved in or witnessed an elephant being harmed	elephant_harmed_bar_chart
28	Yes/No	Report conflict with elephants	report_elephant_conflict_bar_chart
29	Yes/No	Interested in a community dialogue	elephant_community_dialogue_bar_chart
30	True/False	Male with facial secretion may be more aggressive	male_elephant_secretion_bar_chart
31	True/False	Raised trunk means aware of a person	elephant_human_awareness_bar_chart
32	True/False	Head shaking means annoyed	elephant_shake_head_bar_chart
33	True/False	Can smell and hear from far away	elephant_smell_far_away_bar_chart
34	True/False	Move seasonally for food and water	elephants_move_seasonally_bar_chart
35	True/False	Females are protective of their young	female_elephants_protective_bar_chart
36	True/False	Females live in family groups	female_elephants_family_groups_bar_chart
37	True/False	Involved in or witnessed an elephant being harmed	witness_elephant_harm_bar_chart

Note: Widget 37 charts the same question as widget 27. It sits in the True/False group, but the answers are Yes/No.

6. Demographic Table

Task: **format_demographic_table**, run on the binned table with these columns of interest:

- Gender of Participant
- What is your Age
- What is your Tribe
- How many People are there in your Household
- What is your Highest Level of Completed Education

The result is saved with **persist_df** (filetype: geoparquet) as **demographic-data.parquet**. The Word report reads it back from the results directory by that exact name.

7. Likert Chart & Elephant Sentiment Score

7.1 Statements used

retain_agree_disagree keeps id, time, serial_number, geometry, five demographic columns (age group, education, tribe, marital status, gender), household size, age, and the nine statements below. The two 'I Receive Benefits...' statements are relabelled in section 3.5 but are **not** used here.

Direction	Statement
Negative	Elephants Hurt Members of my Community
Negative	Elephants should only Live Inside the Parks
Negative	It is Acceptable to Harm Elephants if they have Damaged Physical Property or Livestock
Negative	It is Acceptable to Harm Elephants if they have Hurt People
Negative	Elephants have a Negative Effect on my Income/Livelihood
Negative	Elephants Negatively Impact my Emotional Wellbeing
Positive	Elephants are Important for a Healthy Ecosystem
Positive	It is Important for Elephants to Live in this Area
Positive	It is Possible for both Humans and Elephants to Live Together in this Area

7.2 Likert chart (widget 38)

Parameter	Value
Task	draw_likert_chart
response_order	Strongly Disagree → Disagree → Neutral → Agree → Strongly Agree
Colours	Strongly Disagree #2c5282, Disagree #4299e1, Neutral / No Response / I Don't Know #a0aec0, Agree #ed8936, Strongly Agree #c05621

Parameter	Value
show_percentage / sort_questions / sort_by	true / false / positive
Output	ele_sentiment_chart.html / .png

7.3 Scoring

Answer	Negative statements	Positive statements
Strongly Disagree	5	1
Disagree	4	2
Neutral	3	3
Agree	2	4
Strongly Agree	1	5
I Don't Know / No Response	0	0

Higher scores always mean a more positive attitude towards elephants. The steps that follow:

Task id	Task	Result
replace_negative / replace_positive	replace_column_values	Answers → scores (table above)
convt_ele_int	convert_columns_to_int	Scores as int (coerce, fill 0)
create_sentiment_score	apply_arithmetic_operation_over_rows (ext-mnc)	sentiment_score = sum of the 9 scores
sentiment_score_mean	apply_arithmetic_operation_over_columns	sentiment_score_mean = mean of the 9 scores
round_sentiment_score	round_column_values	Rounded to 1 decimal place
sentiment_bins	apply_classification	sentiment_bins: equal_interval, k = 5, range labels, 0 dp
replace_bins_mapping	replace_column_values	0 - 1 → Strongly Disagree, 1 - 2 → Disagree, 2 - 3 → Neutral, 3 - 4 → Agree, 4 - 5 → Strongly Agree

Note: 'I Don't Know' and 'No Response' score 0 and are **included** in the mean, so respondents who skip statements are pulled towards the low (negative) end of the scale.

Note: The equal-interval bins are fitted to the observed range of sentiment_score_mean, not a fixed 0–5 scale. The category mapping only matches when the rounded bin labels come out as 0 - 1 ... 4 - 5. Any other label is left unmapped and appears grey on the attitude map.

The scored table (after category mapping) is saved as **elephant-sentiment-table.gpkg**.

8. Statistical Comparisons

Every comparison uses `sentiment_score_mean` from the `sentiment_bins` output.

8.1 Boxplots

Task: `ecoscope_workflows_ext_big_life.tasks.results.draw_boxplot` (vertical, boxmode group, no points, colour #7eb0d5).

Widget	x-axis	Output file
39	Gender of Participant	gender_boxplot_chart
40	What is your Age Group	age_group_boxplot_chart
41	What is your Highest Level of Completed Education	education_level_boxplot_chart
42	What is your Tribe	tribe_boxplot_chart
43	What is your marital status	marital_status_boxplot_chart

8.2 Tukey HSD

For each factor, `filter_df` removes 'No Response'. `compute_tukey_comparisons` (confidence_level 0.95) then runs the pairwise test and `ecoscope_workflows_ext_ate.tasks.results.draw_tukey_chart` plots the mean differences (significant #ff6b6b, non-significant #4ecdc4).

Widget	Group column	Output file
44	Gender of Participant	gender_tukey_chart
45	What is your marital status	marital_status_tukey_chart
46	What is your Age Group	age_group_tukey_chart
47	What is your Highest Level of Completed Education	education_tukey_chart

Note: A Tukey comparison needs at least two groups with more than one respondent each. With a narrow time range this can fail or skip.

8.3 Scatter plots with OLS trendline

Task: `ecoscope_workflows_ext_ate.tasks.results.draw_scatter_chart`, markers only, with a red solid OLS trendline (width 2).

Widget	x-axis	Output file
48	How many People are there in your Household	household_sentiment_scatter_chart
49	What is your Age	age_sentiment_scatter_chart

Note: The previous version's Type II ANOVA table (`anova_results.csv`) is no longer produced.

9. Maps

9.1 Static context layers

Built with `ecoscope_workflows_ext_custom.tasks.results.create_geojson_layer` after reprojecting to EPSG:4326. `combine_deckgl_map_layers` adds all three to every map as static layers.

Layer	Style	Legend
Land use (apply_color_map on land_use)	filled + stroked, line width 0.85, opacity 0.75. Rangeland #a39c91, Rangeland and settlement #ffa07a, Settlement and agriculture #2f4f4f, Conservancies #8fbc8b, National parks and reserves #4d6600	Map Layers (5 entries)
Ranch boundaries	outline only, black, width 0.95, opacity 0.55	Ranch boundaries
Electric fence	outline only, #2d62b1, width 5.5, opacity 1.0	Electric fence

9.2 Viewport

Task group **Map Zoom & Extent**: `envelope_gdf` (expansion_factor 1.05) is applied to the **ranch boundaries** layer, and `compute_view_state_from_gdf` (pitch 0, bearing 0, max_zoom 15) builds the view. The viewport is therefore always Amboseli, not the extent of the survey points. All three maps use `draw_map` with max_zoom 10, static: false, and the legend at bottom-right.

9.3 Point layers

Each map adds one `create_scatterplot_layer` built from the `replace_bins_mapping` table (radius 4, opacity 0.75, black outline 0.25).

Widget	Map	Colouring	Output file
50	Survey events	Single colour [220, 20, 60] (#dc143c), legend 'Survey Location'	survey_event_map
51	Gender	map_column_value on Gender of Participant: Female #e9967a, Male #0000ff, No Response / other #cfcfc4; add_rgba_columns_from_hex → gender_colors_rgba	gender_event_map
52	Attitude	map_column_value on sentiment_bins: Strongly Disagree #dc143c, Disagree #d2691e, Neutral #ffff00, Agree #7fff00, Strongly Agree #0000ff, other #cfcfc4 → sentiment_colors_rgba	sentiment_event_map

10. PNG Conversion, Word Report & Dashboard

10.1 PNG conversion

Task id	Inputs	Config
convert_charts_to_png	All 50 chart HTML files (one call)	1280 × 720, device_scale_factor 2.0, wait_for_timeout 1 ms, max_concurrent_pages 10
convert_survey_map_png convert_gender_map_png convert_sent_map_png	/ One map each / /	device_scale_factor 2.0, wait_for_timeout 40 000 ms (tile loading), max_concurrent_pages 1

Each PNG is written to ECOSCOPE_WORKFLOWS_RESULTS with the same stem as its HTML file.

10.2 Word report

Task: **generate_survey_report** (ecoscope-workflows-ext-ate). The Dropbox template is rendered with docxtpl:

- **Images:** every .png/.jpg/... found anywhere under the results directory is exposed as a template variable named after its file stem (e.g. `{{ what_is_your_age_group_pie_chart }}`), inserted at 11.11 × 6.5 cm.
- **Demographics:** demographic-data.parquet (or .csv) is read into **demographics** and **total_responses**.
- **Metadata:** report_period (since–until in the chosen time format), time_generated, and prepared_by = 'Ecoscope'.

The output is **social_survey_report.docx**.

Note: In the task graph, generate_survey_doc depends only on the template download and the time range, **not** on the PNG conversion tasks. The report only contains the images that exist when it runs, so run the workflow sequentially (or make the report depend on the PNG tasks) to guarantee a complete report. Renaming a chart's filename in spec.yaml also breaks its template tag.

10.3 Dashboard

Charts are wrapped with **create_plot_widget_single_view** (widgets 0–49) and maps with **create_map_widget_single_view** (widgets 50–52). **gather_dashboard** combines the 53 widgets with the workflow details, time range, and groupers. The grid positions live in **layout.json** (53 entries, 10 × 10 units each).

11. Output Files

All outputs are written to **ECOSCOPE_WORKFLOWS_RESULTS**. Every chart and map exists as .html and .png. Sections 4, 5, 8 and 9 list the individual chart file names.

File	Description
survey-events.gpkg	Cleaned, relabelled survey responses
demographic-data.parquet	Demographic summary table (used by the report)
elephant-sentiment-table.gpkg	Statement scores, sentiment_score, sentiment_score_mean, sentiment_bins
*_pie_chart.html / .png (×11)	Pie charts
*_bar_chart.html / .png (×27)	Bar charts
ele_sentiment_chart.html / .png	Likert chart
*_boxplot_chart.html / .png (×5)	Sentiment boxplots
*_tukey_chart.html / .png (×4)	Tukey HSD plots
*_sentiment_scatter_chart.html / .png (×2)	Scatter plots with OLS trendline
survey_event_map / gender_event_map / sentiment_event_map (.html / .png)	Maps
amboseli_*.gpkg (×3), ate_survey_template_280857_updated.docx	Downloaded inputs
social_survey_report.docx	Final Word report

12. Workflow Execution Logic

12.1 Skip conditions

Every task inherits **task-instance-defaults**. The map scatterplot layers and draw_map tasks also declare the same conditions explicitly:

Condition	Behaviour
any_is_empty_df	Skip the task if any input DataFrame is empty
any_dependency_skipped	Skip the task if any upstream dependency was skipped

A single empty result (for example, no households with dogs) skips that chart, its PNG, and its widget. The rest of the run continues.

12.2 Data flow summary

Stage	Main table	Feeds
Clean & relabel	replace_marital_status	Pie charts, survey-events.gpkg, binning
Bin	order_bins	Binned bar charts, demographic table, retain_* subsets
Subset	retain_yes_no / retain_true_false / retain_agree_disagree	Yes/No bars, True/False bars, Likert chart + scoring
Score	sentiment_bins → replace_bins_mapping	Boxplots, Tukey, scatter / maps, sentiment gpkg
Publish	persist_* → html_to_png → generate_survey_report; widget_* → gather_dashboard	PNG files, Word report, dashboard

13. Software Versions

Package	Version pinned in spec.yaml	Channel
ecoscope-platform	2.18.0	ecoscope-workflows
ecoscope-workflows-ext-custom	0.1.0rc14.*	ecoscope-workflows-custom
ecoscope-workflows-ext-ste	0.0.0rc1.*	ecoscope-workflows-custom
ecoscope-workflows-ext-mnc	1.0.5.*	ecoscope-workflows-custom
ecoscope-workflows-ext-wwf-virunga	0.0.0rc9.*	ecoscope-workflows-custom
ecoscope-workflows-ext-big-life	1.0.2.*	ecoscope-workflows-custom
ecoscope-workflows-ext-ate	1.0.3.*	ecoscope-workflows-custom
pydeck	0.9.2	conda-forge
opentelemetry-sdk	>=1.20.0,<2.0.0	conda-forge

Build tooling (pixi.toml): wt-compiler >=0.8.2,<0.9, graphviz, go-yq. Compiled package version: **ecoscope-workflows-social-survey-workflow/VERSION.yaml**.